ESP32-P4

Series SoC Errata

Version v1.3

ESPRESSIF

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1 Chip Revision Identification

Espressif is introducing a new **vM.X** numbering scheme to indicate chip revisions. This guide outlines the structure of this scheme and provides information on chip errata and additional identification methods.

1.1 Chip Revision Numbering Scheme

The new numbering scheme **vM.X** consists of the major and minor numbers described below.

M –Major number, indicating the major revision of the chip product. If this number changes, it means the software used for the previous version of the product is incompatible with the new product, and the software version shall be upgraded for the use of the new product.

X –Minor number, indicating the minor revision of the chip product. If this number changes, it means the software used for the previous version of the product is compatible with the new product, and there is no need to upgrade the software.

The **vM.X** scheme replaces previously used chip revision schemes, including ECOx numbers, Vxxx, and other formats if any.

1.2 Primary Identification Methods

eFuse Bits

The chip revision is encoded using two eFuse fields:

- EFUSE_RD_MAC_SPI_SYS_2_REG[23]
- EFUSE_RD_MAC_SPI_SYS_2_REG[5:0]

Table 1.1: Chip Revision Identification by eFuse Bits

	eFuse Bit	Chip Revision					
		v0.0	v1.0	v1.3	v3.0	v3.1	v3.2
Major Number	EFUSE_RD_MAC_SPI_SYS_2_REG[23]	0	0	0	0	0	0
	EFUSE_RD_MAC_SPI_SYS_2_REG[5]	0	0	0	1	1	1
	EFUSE_RD_MAC_SPI_SYS_2_REG[4]	0	1	1	1	1	1
Minor Number	EFUSE_RD_MAC_SPI_SYS_2_REG[3]	0	0	0	0	0	0
	EFUSE_RD_MAC_SPI_SYS_2_REG[2]	0	0	0	0	0	0
	EFUSE_RD_MAC_SPI_SYS_2_REG[1]	0	0	1	0	0	1
	EFUSE_RD_MAC_SPI_SYS_2_REG[0]	0	0	1	0	1	0

Chip Marking

- **Manufacturing Code** line in chip marking

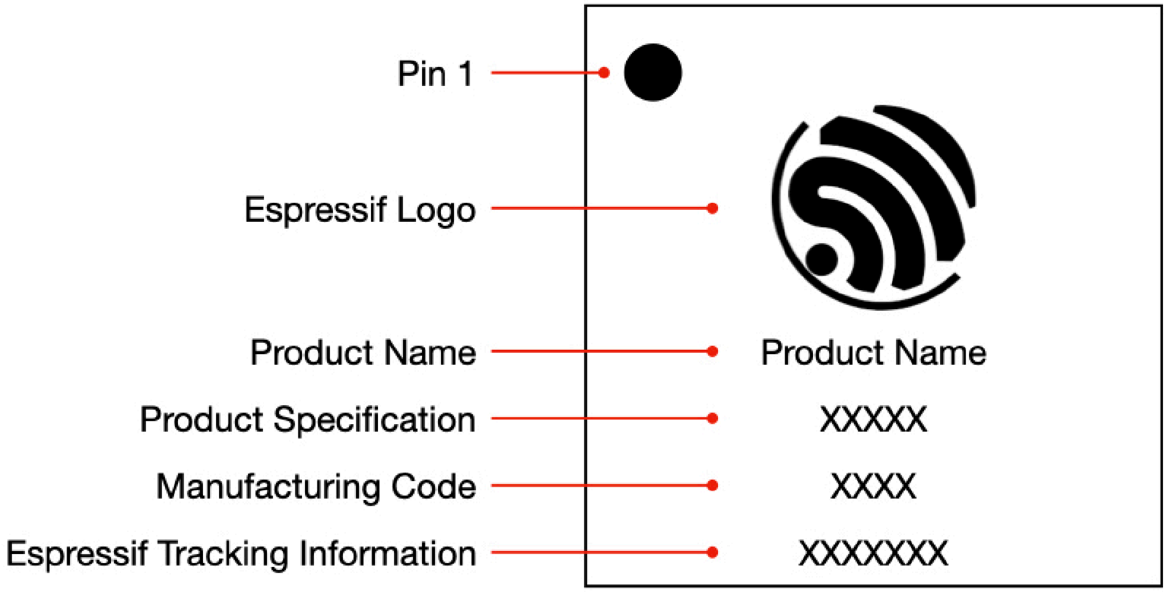

Figure 1.1: Chip Marking Diagram

Table 1.2: Chip Revision Identification by Chip Marking

Chip Revision	Manufacturing Code
v0.0	X A XX
v1.0	X C XX
v1.3	X E XX
v3.0	X F XX
v3.1	X G XX
v3.2	X H XX

1.3 Additional Identification Methods

Date Code

Some errors in the chip product don't need to be fixed at the silicon level, or in other words in a new chip revision.

In this case, the chip may be identified by **Date Code** in chip marking (see *Chip Marking*). For more information, please refer to ESP32-P4 Chip Packaging Information > Chip Silk Marking.

1.4 ESP-IDF Release Compatibility

Information about ESP-IDF release that supports a specific chip revision is provided in Compatibility Between ESP-IDF Releases and Revisions of Espressif SoCs.

1.5 Related Documents

- For more information about the chip revision upgrade and their identification of series products, please refer to ESP32-P4 Product/Process Change Notifications (PCN).
- For more information about the chip revision numbering scheme, see Compatibility Advisory for Chip Revision Numbering Scheme.

2 Errata Summary

Table 2.1: Errata summary

Category	Errata No.	Descriptions	Affected Revisions					
			v0.0	v1.0	v1.3	v3.0	v3.1	v3.2
RMT	RMT-176	*[RMT-176] The Idle State Signal Level Might Run into Error in RMT Continuous TX Mode*	Y	Y	Y			
I2C	I2C-308	*[I2C-308] I2C Slave Fails in Multiple-read Under Non-FIFO Mode*	Y	Y	Y			
MSPI	MSPI-749	*[MSPI-749] Load Access Fault During Chip Power-on or Deep-Sleep Wake-up*				Y		
	MSPI-750	*[MSPI-750] PSRAM Unaligned DMA Read Operations May Return Old Data When Accessing Overlapping Addresses*				Y		
	MSPI-751	*[MSPI-751] Data Errors Caused by Asynchronous Timing Issues in the MSPI Address Overlap Detection Function When Read/Write Operations Overlap at Specific Frequencies*				Y		
ROM	ROM-764	*[ROM-764] Secure Boot Verification Failure Caused by Incorrect Buffer Address in ROM*				Y		
	ROM-770	*[ROM-770] Secure Download Mode Flash Power-On Failure*					Y	
	ROM-816	*[ROM-816] Device Hang When Flash Power-On Sequence Runs Twice with rom_download_xpd_on eFuse*						Y
Analog	Analog-765	*[Analog-765] Output Regulators Cannot Generate a Reliable Supply When Peripheral Power Domain Is Off*				Y		
DMA	DMA-767	*[DMA-767] DMA Channel 0 Transaction ID Overlap Causes Permission Management Issue*				Y		
APM	APM-560	*[APM-560] Unauthorized AHB Access May Block Subsequent PSRAM or Flash Transactions*	Y	Y	Y	Y		
ECDSA_DS [1]	ECDSA_DS-836	*[ECDSA_DS-836] Signatures with Invalid r and s Values Are Incorrectly Accepted*				Y	Y	Y
	ECDSA_DS-837	*[ECDSA_DS-837] Signatures with Invalid s Values Are Incorrectly Accepted*	Y	Y	Y			

[1] ECDSA_DS: ECDSA Digital Signature Peripheral.

3 All Errata Descriptions

3.1 [RMT-176] The Idle State Signal Level Might Run into Error in RMT Continuous TX Mode

Affected revisions: v0.0 v1.0 v1.3

Description

In ESP32-P4's RMT module, if the continuous TX mode is enabled, it is expected that the data transmission stops after the data is sent for RMT_TX_LOOP_NUM_CHn rounds, and after that, the signal level in idle state should be controlled by the "level" field of the end-marker.

However, in real situation, after the data transmission stops, the channel's idle state signal level is not controlled by the "level" field of the end-marker, but by the level in the data wrapped back, which is indeterminate.

Workarounds

Users are suggested to set RMT_IDLE_OUT_EN_CHn to 1 to only use registers to control the idle level.

This issue has been bypassed since the first ESP-IDF version that supports continuous TX mode (v5.2). In these versions of ESP-IDF, it is configured that the idle level can only be controlled by registers.

Solution

Fixed in chip revision v3.0.

3.2 [I2C-308] I2C Slave Fails in Multiple-read Under Non-FIFO Mode

Affected revisions: v0.0 v1.0 v1.3

Description

If the I2C slave non-FIFO mode is enabled and the master performs multiple-read operation on the slave, the master can not correctly read the data from the slave.

Workarounds

Set I2C_FIFO_ADDR_CFG_EN and I2C_SLV_TX_AUTO_START_EN to 1 and I2C_FIFO_PRT_EN to 0 for the slave.

After configuration, the master must access the slave using the following command sequence: RSTART -> WRITE (slave addr, fifo addr) -> RSTART -> WRITE (slave addr) -> READ (NACK) -> STOP, with only one byte allowed to be read each time.

Solution

Fixed in chip revision v3.0.

3.3 [MSPI-749] Load Access Fault During Chip Power-on or Deep-Sleep Wake-up

Affected revisions: v3.0

Description

During the power-on or wake-up process of ESP32-P4, due to abnormal handling inside the MSPI IP read-request channel, the MSPI may fail to correctly process the first and second access requests initiated by the AXI system bus. This can occasionally result in unexpected error responses, causing the boot process to fail.

Workarounds

Power-on sequence: The power-on sequence is fixed. If the sequence fails, the system can only rely on the watchdog timeout to reset the chip and perform a "second power-on". After this second power-on, the flash MSPI module can resume normal operation.

Wake-up sequence: During the Deep-sleep wake-up, the LP memory region remains powered and retains its contents. A small executable program can be pre-loaded into LP memory before entering sleep. This program should perform two dummy read accesses. Upon wake-up, the CPU will execute this program first and then jump to the ROM code to continue the normal boot process. The code is as follows:

```
REG32_WR(0x500ca000, 0x23);
CLEAR_PERI_REG_MASK(0x5008c03c, 0x80000000);

// Disable CPU error response handling
SET_PERI_REG_MASK(0x500e51a4, 0x00000007);

// Enable AXI interface
REG32_WR(0x500ca000, 0x23);
SET_PERI_REG_MASK(0x5008c03c, 0x00000001);
```

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```
// Configure one MSPI MMU entry to map the AXI address to the flash address
REG32_WR(0x500ca000, 0x23);
SET_PERI_REG_MASK(0x5008c380, 0x0);
REG32_WR(0x500ca000, 0x23);
SET_PERI_REG_MASK(0x5008c37c, 0x1000);

//Perform the first and second AXI read accesses from flash MSPI
REG32_RD(0x80000000);
REG32_WR(0x500ca000, 0x23);
REG32_RD(0x80000040);
REG32_WR(0x500ca000, 0x23);

//Re-enable CPU error response handling
CLEAR_PERI_REG_MASK(0x500e51a4, 0x00000007);

SET_PERI_REG_MASK(0x50111014, 0x8000);
SET_PERI_REG_MASK(0x50111014, 0x2000);
```

PSRAM MSPI: During PSRAM MSPI initialization, inserting two dummy reads before the first actual AXI read access can effectively avoid the initial access anomaly.

Solution

Fixed in chip revision v3.1.

3.4 [MSPI-750] PSRAM Unaligned DMA Read Operations May Return Old Data When Accessing Overlapping Addresses

Affected revisions: v3.0

Description

When accessing PSRAM randomly through DMA or CACHE, if a write operation to a certain address range is followed by a read operation to the same range, data readout errors may occur. The triggering conditions are:

- Any write request is executed.
- The read operation is a DMA-initiated read request with a burst size of 1 byte or 2 bytes, and the access address is not 4-byte aligned.
- The address range accessed by the read operation overlaps with the address range previously written.

Under these conditions, the read operation may fail to obtain the new data written by the prior write operation and instead return old data that existed in PSRAM before the write.

Root cause: The MSPI IP's internal "address overlap detection" mechanism has an anomaly when processing the above unaligned read requests. The computed address range used for overlap detection may be 1 to 3 bytes shorter at the end than the actual address range the read request should access. This defect may cause the overlap detection to fail, resulting in incorrect data reads.

Workarounds

It is recommended that software applications enforce 4-byte alignment for all masters that may issue unaligned accesses (for example, burst sizes of 1 or 2 bytes), such as USB or SDMMC. This prevents the issue from occurring.

Solution

Fixed in chip revision v3.1.

3.5 [MSPI-751] Data Errors Caused by Asynchronous Timing Issues in the MSPI Address Overlap Detection Function When Read/Write Operations Overlap at Specific Frequencies

Affected revisions: v3.0

Description

When accessing PSRAM randomly through DMA or CACHE, if a write operation to a certain address range is followed by a read operation to the same range, data readout errors may occur under specific clock frequency configurations. The triggering conditions are:

- During random PSRAM accesses through DMA or CACHE, a "write followed by read" sequence occurs on the same address range.
- The frequency ratio between the AXI bus (freq_axi) and the MSPI core (freq_core) exceeds the safe range. The exact safe range depends on whether AXI concatenation is enabled and the actual timing characteristics of the chip.

Under these conditions, the read operation may fail to obtain the new data written by the prior write operation and instead return old data that existed in PSRAM before the write. This error breaks data consistency and may impact normal chip functionality.

Root cause: The MSPI IP's internal "address overlap detection" function has an asynchronous timing issue. When the clock-frequency ratio is near a critical boundary, this timing issue may cause the detection logic to fail, preventing it from correctly identifying overlapping addresses and eventually leading to incorrect data reads.

Workaround

To ensure system stability, software configurations must satisfy the following frequency constraints, based on whether AXI concatenation is enabled and the timing conditions:

Timing Condition	AXI Concatenation Status	Frequency Constraint
Assuming poor timing	Enabled	3 × freq_core ≥ freq_axi
	Disabled	freq_core ≥ freq_axi
Assuming normal timing	Enabled	4 × freq_core ≥ freq_axi
	Disabled	2 × freq_core ≥ freq_axi

Solution

Fixed in chip revision v3.1.

3.6 [ROM-764] Secure Boot Verification Failure Caused by Incorrect Buffer Address in ROM

Affected revisions: v3.0

Description

In the ROM, the start address of the secure-boot stage buffer is incorrectly configured. It points to the L2 cache data memory (0x4FF0_0000), which is part of the L2 cache and not writable by the CPU.

Because of this misconfiguration, data stored in the secure-boot buffer may be lost, potentially causing hash calculation failures or signature verification failures during the secure-boot process.

Workaround

Do not enable secure boot.

Solution

Fixed in chip revision v3.1.

3.7 [ROM-770] Secure Download Mode Flash Power-On Failure

Affected revisions: v3.1

Description

To support both 1.8 V and 3.3 V flash, ESP32-P4 introduces a power-control mechanism in which, in Joint Download Boot Mode, the PMU does not power on the flash by default. In the normal download flow, software determines the flash operating voltage, updates eFuse PXA0_TIEH_SEL_0, and configures the related registers (this sequence is integrated into the esptool download command) to power on the flash before programming.

However, the ROM includes a Secure Download feature. When this feature is enabled via the eFuse ENABLE_SECURITY_DOWNLOAD, the ROM rejects register read/write commands and only allows flash programming operations. As a result, software has no opportunity to configure the flash power-control registers, preventing the flash from being powered on and causing the download process to fail.

Workaround

- Do not enable Secure Download Mode.
- Alternatively, provide a board-level mechanism to switch between internal and external flash power to ensure the flash is powered during download. Note that continuous external flash power may affect some IDF features, for example, the flash may not be able to power down in sleep mode. Therefore, it is recommended to switch back to internal flash power before entering sleep mode.

Solution

Fixed in chip revision v3.2.

3.8 [ROM-816] Device Hang When Flash Power-On Sequence Runs Twice with `rom_download_xpd_on` eFuse

Affected revisions: v3.2

Description

To address the issue that, in Secure Download Mode on ESP32-P4 chip revision v3.1 and earlier versions, esptool cannot power on the flash, ESP32-P4 chip revision v3.2 introduces a new eFuse `rom_download_xpd_on`. When this eFuse bit is programmed, the ROM automatically runs the flash power-on sequence during the download flow.

This power-on sequence must not run twice within the same power-on reset (POR) window; otherwise the ROM flow may fail to exit and the chip can hang.

Situations that can lead to the sequence running twice include:

- The `rom_download_xpd_on` eFuse is already programmed, but the host still sends a flash power-on sequence to the chip. A fixed esptool checks this eFuse bit when not in Secure Download Mode and only sends the automatic flash power-on sequence if the bit is not programmed, so this case is normally avoided.

- In Secure Download Mode, `rom_download_xpd_on` is programmed through the USB UART interface, followed by USB UART force download to flash firmware. Because a reset from USB UART is typically not a full POR, entering download again after reset can be equivalent to executing the flash power-on path twice within the same power-on context, which triggers the hang risk described above.

Workaround

- Program the `rom_download_xpd_on` bit only when Secure Download Mode is actually required.
- If you use the USB UART interface for firmware download, perform a full POR before entering download after programming this eFuse bit.

Solution

No fix scheduled.

3.9 [Analog-765] Output Regulators Cannot Generate a Reliable Supply When Peripheral Power Domain Is Off

Affected revisions: v3.0

Description

The output regulators can not generate a reliable supply when the peripheral power domain is turned off.

Workaround

If you intend to use output regulators as the supply source for external PCB components, do not turn off the peripheral power domain in Light-sleep mode.

Solution

Fixed in chip revision v3.1.

3.10 [DMA-767] DMA Channel 0 Transaction ID Overlap Causes Permission Management Issue

Affected revisions: v3.0

Description

When performing a memory-to-memory transfer using AHB DMA channel 0, the transfer is assigned transaction ID "a", which is identical to the transaction ID used by the RMT peripheral DMA.

Because of this ID overlap, memory-to-memory transfers on channel 0 share the same permission settings as the RMT peripheral and cannot be controlled independently. This may lead to unintended or unauthorized accesses in systems requiring isolated permission management.

Workaround

Avoid using memory-to-memory transfer channel 0. Perform all memory-to-memory transfers using channel 1 or channel 2.

Solution

Fixed in chip revision v3.1.

3.11 [APM-560] Unauthorized AHB Access May Block Subsequent PSRAM or Flash Transactions

Affected revisions: v0.0 v1.0 v1.3 v3.0

Description

When multiple AHB Masters (USB OTGFS, USB OTGHS, GMAC, SDMMC, Trace0/1, AHB PDMA, L2MEM Monitor, TCM Monitor, etc.) concurrently access PSRAM or flash, and one of the masters does not have access permission, the access permission check in the DMA APM correctly intercepts the unauthorized transaction.

During interception, the DMA APM does not properly mask downstream responses, so it leads to abnormal subsequent transfers. As a result, after an unauthorized access, even valid accesses initiated by an authorized AHB Master may be stalled.

Workaround

Avoid triggering unauthorized accesses to PSRAM or flash. If unauthorized access cannot be avoided and the system enters the stuck state described above, the only available recovery method is to perform a system reset.

Solution

Fixed in chip revision v3.1.

3.12 [ECDSA_DS-836] Signatures with Invalid `r` and `s` Values Are Incorrectly Accepted

Affected revisions: v3.0 v3.1 v3.2

Description

When the signature `{r = 0 or n, s = 0 or n}` with invalid `r` and `s` values is submitted to the ECDSA_DS peripheral against any message and public key, the peripheral incorrectly reports the signature as "valid".

Workarounds

Use RSA_DS Secure Boot instead of ECDSA_DS Secure Boot.

Solution

No fix scheduled.

3.13 [ECDSA_DS-837] Signatures with Invalid `s` Values Are Incorrectly Accepted

Affected revisions: v0.0 v1.0 v1.3

Description

When the signature `{r = (Qa + G).x, s = 0 or n}` with an invalid `s` value is submitted to the ECDSA_DS peripheral against any message and public key, the peripheral incorrectly reports the signature as "valid".

Workarounds

Use RSA_DS Secure Boot instead of ECDSA_DS Secure Boot.

Solution

No fix scheduled.

4 Revision History

Table 4.1: Revision History

Date	Version	Release Notes
2026-07-10	v1.3	· *Errata Summary* – Improved formatting · *All Errata Descriptions* – Added Section *[ECDSA_DS-836] Signatures with Invalid r and s Values Are Incorrectly Accepted* – Added Section *[ECDSA_DS-837] Signatures with Invalid s Values Are Incorrectly Accepted*
2026-04-20	v1.2	· *Chip Revision Identification* – Added information about chip revisions v3.2 · *All Errata Descriptions* – In Section *[ROM-770] Secure Download Mode Flash Power-On Failure*, updated solution to "fixed in chip revision v3.2" – Added Section *[ROM-816] Device Hang When Flash Power-On Sequence Runs Twice with rom_download_xpd_on eFuse*
2026-02-12	v1.1	· *Chip Revision Identification* – Added information about chip revisions v3.0 and v3.1 · *All Errata Descriptions* – Added Section *[MSPI-749] Load Access Fault During Chip Power-on or Deep-Sleep Wake-up* – Added Section *[MSPI-750] PSRAM Unaligned DMA Read Operations May Return Old Data When Accessing Overlapping Addresses* – Added Section *[MSPI-751] Data Errors Caused by Asynchronous Timing Issues in the MSPI Address Overlap Detection Function When Read/Write Operations Overlap at Specific Frequencies* – Added Section *[ROM-764] Secure Boot Verification Failure Caused by Incorrect Buffer Address in ROM* – Added Section *[Analog-765] Output Regulators Cannot Generate a Reliable Supply When Peripheral Power Domain Is Off* – Added Section *[DMA-767] DMA Channel 0 Transaction ID Overlap Causes Permission Management Issue* – Added Section *[APM-560] Unauthorized AHB Access May Block Subsequent PSRAM or Flash Transactions* – Added Section *[ROM-770] Secure Download Mode Flash Power-On Failure*
2025-07-08	v1.0	First release

5 Related Documentation and Resources

5.1 Related Documentation

- ESP32-P4 Datasheet –Specifications of the ESP32-P4 hardware.
- ESP32-P4 Technical Reference Manual –Detailed information on how to use the ESP32-P4 memory and peripherals.
- ESP32-P4 Hardware Design Guidelines –Guidelines on how to integrate the ESP32-P4 into your hardware product.
- Certificates
  https://espressif.com/en/support/documents/certificates
- ESP32-P4 Product/Process Change Notifications (PCN)
  https://espressif.com/en/support/documents/pcns?keys=ESP32-P4
- ESP32-P4 Advisories –Information on security, bugs, compatibility, component reliability.
  https://espressif.com/en/support/documents/advisories?keys=ESP32-P4
- Documentation Updates and Update Notification Subscription
  https://espressif.com/en/support/download/documents

5.2 Developer Zone

- ESP-IDF Programming Guide for ESP32-P4 –Extensive documentation for the ESP-IDF development framework.
- ESP-IDF and other development frameworks on GitHub.
  https://github.com/espressif
- ESP32 BBS Forum –Engineer-to-Engineer (E2E) Community for Espressif products where you can post questions, share knowledge, explore ideas, and help solve problems with fellow engineers.
  https://esp32.com/
- The ESP Journal –Best Practices, Articles, and Notes from Espressif folks.
  https://blog.espressif.com/
- See the tabs SDKs and Demos, Apps, Tools, AT Firmware.
  https://espressif.com/en/support/download/sdks-demos

5.3 Products

- ESP32-P4 Series SoCs –Browse through all ESP32-P4 SoCs.
  https://espressif.com/en/products/socs?id=ESP32-P4
- ESP32-P4 Series Modules –Browse through all ESP32-P4-based modules.
  https://espressif.com/en/products/modules?id=ESP32-P4
- ESP32-P4 Series DevKits –Browse through all ESP32-P4-based devkits.
  https://espressif.com/en/products/devkits?id=ESP32-P4

- ESP Product Selector –Find an Espressif hardware product suitable for your needs by comparing or applying filters.
  https://products.espressif.com/#/product-selector

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